SMALL MAXIMAL INTEGER SIDON SETS BY FINITE PHASE COLORINGS

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ABSTRACT. We prove that every sufficiently large integer interval $\{1, \dots, N\}$ contains an inclusion-maximal Sidon set of cardinality $O(N^{1/3})$. The construction combines a quadratic Sidon seed with a periodically labeled family of anchors. Finite-dimensional bounds for affine sign patterns produce a fixed alphabet of integer translates and a periodic word that covers every relevant local window. A pair of separated alphabets supplies Sidon, sum-free tags for every prime, so the prime-selection step uses only Bertrand's postulate. The resulting seed blocks every point outside two intervals of length $O(p^2)$ and one residue class, where $p \asymp N^{1/3}$; a maximal completion adds only $O(p)$ elements. A self-contained Lean file verifies the upper bound, including the uniform pattern counts and the placement argument, against the formal statement of Erdős problem 156.

1. INTRODUCTION

A set of integers A is a *Sidon set* if

$$a + b = c + d, \qquad a, b, c, d \in A,$$

implies $\{a, b\} = \{c, d\}$ as multisets. In particular, diagonal sums $2a$ are included in the definition. A Sidon subset of $[1, N] \cap \mathbb{Z}$ is *maximal* if adjoining any further point of that interval destroys the Sidon property. Write

$$s(N) = \min\{|A| : A \subseteq [1, N] \cap \mathbb{Z} \text{ is a maximal Sidon set}\}.$$

Maximality here is with respect to inclusion, rather than maximum cardinality.

Erdős, Sárközy, and Sós studied this problem in their work on sumsets of Sidon sets [1]. Ruzsa [2] proved the existence of maximal Sidon sets of size $O((N \log N)^{1/3})$. We establish the following upper bound by a construction whose uniformity statements are included in a Lean proof.

Theorem 1.1. *There are absolute constants $C > 0$ and $N_0 \in \mathbb{N}$ such that, for every integer $N \geq N_0$, there is a maximal Sidon set $A \subseteq [1, N] \cap \mathbb{Z}$ satisfying*

$$|A| \leq CN^{1/3}.$$

Equivalently, $s(N) = O(N^{1/3})$.

The exponent is forced by a counting argument. We recall it to distinguish the extremal scale from the construction needed to attain it.

Proposition 1.2. *Every maximal Sidon set $A \subseteq [1, N] \cap \mathbb{Z}$ satisfies $|A| \geq (N/3)^{1/3}$.*

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Proof. The case $N = 0$ is immediate. Otherwise put $m = |A| \geq 1$. For every $x \in [1, N] \setminus A$, a nontrivial equality of pair sums in $A \cup \{x\}$ must take one of the forms

$$x + a = b + c \quad \text{or} \quad 2x = a + b, \qquad a, b, c \in A.$$

Indeed, equalities using x once on each side cancel to a trivial equality. Consequently the interval is contained in the union of A, $A + A - A$, and the set of integral midpoints of pairs from A. Thus $N \leq m + m^3 + m^2 \leq 3m^3$, since $m \geq 1$. □

Together, Theorem 1.1 and Proposition 1.2 give $s(N) \asymp N^{1/3}$. The companion certificate proves the upper bound in Theorem 1.1; the elementary lower-bound argument is included here for context.

Organization and notation. For an integer z and a positive integer q, let $\langle z \rangle_q \in \{0, \ldots, q-1\}$ denote its least nonnegative residue. For a real number u, write $\{u\} = u - \lfloor u \rfloor$. All intervals below are intersected with $\mathbb{Z}$ when used as sets of marks.

We first work in $[0, N)$ and translate by 1 at the end. A quadratic seed C_p supplies controlled differences. A finite alphabet of translates covers all nonmultiples of p in a long interior interval. Two versions of this alphabet produce tags that preserve Sidonicity. A periodic choice of labels then makes a family of anchors cover the interior of the host interval. Only two short boundary intervals and the multiples of p remain available to a maximal completion.

2. Finite pattern selection

The construction uses finite families of patterns even though the arithmetic parameters range over infinitely many primes and targets. The following deliberately nonoptimal estimate agrees with the bound used in the certificate.

Lemma 2.1 (Affine sign patterns)**.** *Let $f_1, \ldots, f_q$ be affine real functions on $\mathbb{R}^d$. The number of joint sign patterns, with each sign in $\{-, 0, +\}$, is at most*

$$\big((d+2)(q+1)^{d+1}\big)^2. \tag{2.1}$$

The same bound applies to any finite-valued label determined by these signs.

Proof. Homogenize the affine functions on $\mathbb{R}^{d+1}$. The family of their positive supports has VC dimension at most $d + 1$. To see the dimension bound, suppose a collection of the corresponding linear functionals is shattered. A nonzero linear dependence among them is impossible: choose a realizing vector on which precisely the functionals with positive coefficients are positive, changing the sign of the dependence first if necessary. Evaluating the dependence then gives a strictly positive sum.

The Sauer–Shelah bound [3, 4] gives at most

$$\sum_{j=0}^{d+1} \binom{q}{j} \leq (d+2)(q+1)^{d+1}$$

positive supports. Apply the same estimate to negative supports. The pair consisting of the positive and negative support determines all zero signs as well, giving (2.1). □

Lemma 2.2 (Fractional-part tests)**.** *Suppose a label is determined by t comparisons*

$$\{f_i(v)\} < g_i(v), \qquad 1 \leq i \leq t,$$

where f_i, g_i are affine functions of $v \in \mathbb{R}^d$, and $|f_i(v)| \leq B$ on the parameter set under consideration, with B a nonnegative integer. Then the number of labels is at most

$$\big((d+2)(2t(2B+1)+1)^{d+1}\big)^2.$$

No bound on the values of g_i is required.

Proof. For each integer $k \in [-B, B]$, record the signs of $f_i - k$ and $f_i - g_i - k$. The first collection determines $\lfloor f_i \rfloor$, including the case that f_i is an integer; the second then determines whether $f_i - \lfloor f_i \rfloor < g_i$. There are $2t(2B+1)$ affine functions. Apply Lemma 2.1. □

Lemma 2.3 (A common hitting word)**.** *Fix $b \geq 2$. For each m, let $\mathcal{G}_m$ be a family of matrices*

$$(G_0, \dots, G_{m-1}), \qquad \varnothing \neq G_i \subseteq \{1, \dots, b\}.$$

If $|\mathcal{G}_m| \leq Am^B$ for fixed constants A, B, then for all sufficiently large m there is a word $w \in \{1, \dots, b\}^m$ that meets every matrix: for each $G \in \mathcal{G}_m$, some i satisfies $w_i \in G_i$. One may also require this property simultaneously for every cyclic rotation of w.

Proof. Choose the letters independently and uniformly. A fixed matrix is missed with probability at most $(1 - 1/b)^m$. The union bound is less than 1 once $Am^B(1 - 1/b)^m < 1$. To include all rotations, use the bound $Am^{B+1}(1 - 1/b)^m$, which also tends to zero. □

3. The quadratic seed and exact differences

Let p be an odd prime and put

$$Q_p = 2p^2, \qquad c_p(t) = t + 2p\langle t^2 \rangle_p, \qquad C_p = \{c_p(t) : 0 \leq t < p\}, \qquad D_p = C_p - C_p. \tag{3.1}$$

Lemma 3.1. *The set C_p is Sidon, has p elements, and is contained in $[0, Q_p)$.*

Proof. If $c_p(i) + c_p(j) = c_p(k) + c_p(l)$, reduction modulo $2p$ gives $i + j = k + l$, since both sums lie in $[0, 2p)$. Subtracting this equality and dividing by $2p$ gives equality of the sums of the quadratic residues. In $\mathbb{F}_p$ we therefore have

$$i + j = k + l, \qquad i^2 + j^2 = k^2 + l^2.$$

As p is odd, these equalities imply $ij = kl$. The two pairs are roots of the same monic quadratic, so they agree as unordered pairs in $\mathbb{F}_p$, and hence as integer pairs in $[0, p)$. Distinct parameters give distinct residues modulo p, proving the cardinality statement. Finally,

$$0 \leq c_p(t) \leq (p-1) + 2p(p-1) < 2p^2.$$

□

Lemma 3.2 (Exact lift with two carries)**.** *Let $1 \leq d < p$ and $1 \leq H < p$. Define*

$$t = \langle (H + d^2)(2d)^{-1} \rangle_p, \qquad \sigma = \mathbf{1}_{t<d}, \qquad u = t - d + p\sigma, \qquad \varepsilon = \mathbf{1}_{\langle t^2 \rangle_p < H}.$$

Then $0 \leq u < p$ and

$$c_p(t) - c_p(u) = d - p\sigma + 2pH - \varepsilon Q_p. \tag{3.2}$$

Proof. The definition of σ gives $t - u = d - p\sigma$ and $0 \leq u < p$. In $\mathbb{F}_p$,

$$t^2 - u^2 = 2dt - d^2 = H.$$

The difference of the two least residues lies strictly between $-p$ and p. It is therefore H or $H - p$. The latter alternative occurs exactly when $\langle t^2 \rangle_p < H$. Substitution in (3.1) proves the identity. □

In particular, if $\tau \in \{0, 1\}$ and $k \geq 0$, then

$$d + p\tau + 2p(H + k) = c_p(t) - c_p(u) + p\big(2k + \tau + \sigma\big) + \varepsilon Q_p. \tag{3.3}$$

Thus the six pairs $(s, \varepsilon) \in \{0, 1, 2\} \times \{0, 1\}$ contain a suitable choice in every row. The identity is in $\mathbb{Z}$, not merely modulo Q_p.

4. A fixed covering alphabet

Definition 4.1. A *covering alphabet* consists of a finite label set $\mathcal{L}$ of size $b \geq 2$, integers $\delta_\ell \in [0, M]$, and $\varepsilon_\ell \in \{0, 1\}$, such that:

(i) equality of two δ-pair sums determines the unordered pair of labels;

(ii) for every odd prime p,

$$I_p(M) \setminus p\mathbb{Z} \subseteq D_p + \{p\delta_\ell + \varepsilon_\ell Q_p : \ell \in \mathcal{L}\}, \qquad I_p(M) = [2p(M+2),\, 2p(p-1)). \tag{4.1}$$

An empty interval in (4.1) causes no difficulty.

Proposition 4.2 (Separated covering alphabets)**.** *Covering alphabets exist. Moreover, for every prescribed $G \geq 0$, one can choose a covering alphabet for which*

$$|\delta_i + \delta_j - \delta_k - \delta_l| > G \quad \text{whenever } \{i, j\} \neq \{k, l\} \tag{4.2}$$

as multisets of labels.

Proof. Fix a positive integer λ with $2\lambda - 4 > G$. For $0 \leq i < m$ set

$$a_i = i + 2mi^2, \qquad k_i = \lambda a_i.$$

The sequence (a_i) is Sidon: a pair-sum equality modulo $2m$ first determines the integer sum of the indices, and then the sum of their squares determines the unordered pair. Any selection of pairs $(s_i, \varepsilon_i) \in \{0, 1, 2\} \times \{0, 1\}$ gives

$$\delta_i = 2k_i + s_i, \qquad \varepsilon_i \in \{0, 1\}, \qquad M = 6\lambda m^3 + 2. \tag{4.3}$$

Here $\delta_i \leq M$ and $k_i \leq M$. For distinct unordered index pairs, the a-pair sums differ by at least 1, whereas the difference of the two s-pair sums has absolute value at most 4. This proves (4.2) for every selection. It remains to make one selection that covers every prime and target.

Write a target in (4.1) uniquely as

$$y = d + p\tau + 2pY, \qquad 1 \leq d < p, \quad \tau \in \{0, 1\}.$$

The interval bounds give $M + 2 \leq Y < p$. Hence $H_i = Y - k_i$ lies in $[1, p)$ for every row. Let $r = (2d)^{-1} \in \mathbb{F}_p$ and $u = r(Y + d^2) \in \mathbb{F}_p$. The successful pair in (3.3) is determined by the tests

$$\begin{aligned} \{\langle u \rangle_p / p - k_i \langle r \rangle_p / p\} &< d/p, \\ \{\langle u^2 \rangle_p / p + k_i \langle -2ur \rangle_p / p + k_i^2 \langle r^2 \rangle_p / p\} &< Y/p - k_i/p. \end{aligned} \tag{4.4}$$

The first test determines σ_i, and the second determines ε_i; put $s_i = \tau + \sigma_i$.

Treat the eight normalized quantities appearing in (4.4) as independent parameters in $[0, 1]^8$. This enlarges the pattern family. Since $k_i = O_\lambda(m^3)$, the arguments inside the fractional parts are bounded in absolute value by $O_\lambda(m^6)$. There are $2m$ tests, so Lemma 2.2 gives $O_\lambda(m^{126})$ matrices of successful choices, including both values of τ. Every row of every such matrix is a singleton. The formal bound is

$$200\big(16(3\lambda + 1)^2\big)^{18} m^{126}.$$

By Lemma 2.3, for sufficiently large m there is a fixed word of six choices that meets every matrix. Fix this word and use (4.3). For any p and y in (4.1), a row at which the selected word agrees with its successful choice supplies the exact representation (3.3). All choices and constants are fixed before p and y are given. □

5. Tags for every prime

We next arrange that arbitrary labels can be attached to a Sidon ruler without creating new equalities of pair sums. The modulus below need not be prime.

Proposition 5.1 (Two alphabets suffice)**.** *There exist two covering alphabets $\mathcal{L}_1, \mathcal{L}_2$ and an integer $K \geq 1$ with the following property. For every integer $p \geq 0$, at least one alphabet admits a shift $T \in \{0, \dots, K-1\}$ such that the residues*

$$v_\ell = 4p + T + \delta_\ell + 2p\varepsilon_\ell \pmod{K} \tag{5.1}$$

determine unordered label pairs from their pair sums and are sum-free:

$$v_i + v_j = v_k + v_l \Longrightarrow \{i, j\} = \{k, l\}, \qquad v_i + v_j \neq v_k.$$

Proof. Choose any covering alphabet $\mathcal{L}_1$ with bound M_1. By Proposition 4.2, choose $\mathcal{L}_2$ with separation greater than $4M_1$ and bound M_2. Let b_1, b_2 be their sizes and put

$$K = 4(M_1 + M_2) + b_1^3 + b_2^3 + 1. \tag{5.2}$$

For an alphabet, first consider the unshifted residues $\delta_\ell + 2p\varepsilon_\ell$. A collision of distinct unordered label pairs gives

$$K \mid d + 2pe, \qquad d = \delta_i + \delta_j - \delta_k - \delta_l, \quad e = \varepsilon_i + \varepsilon_j - \varepsilon_k - \varepsilon_l.$$

We have $d \neq 0$, $|d| \leq 2M$, and $|e| \leq 2$. Since $K > 2M$, such a collision must have $e \neq 0$.

Suppose both alphabets have a collision, with differences (d_1, e_1) and (d_2, e_2). Eliminating p gives

$$K \mid e_2 d_1 - e_1 d_2, \qquad |e_2 d_1 - e_1 d_2| \leq 4(M_1 + M_2) < K.$$

Thus $e_2 d_1 = e_1 d_2$. As $1 \leq |e_1| \leq 2$,

$$|d_2| \leq |e_1 d_2| = |e_2 d_1| \leq 4M_1,$$

contrary to the separation of $\mathcal{L}_2$. At least one unshifted alphabet therefore has unique label pair sums modulo K.

For any b residues $u_\ell \pmod{K}$, translation by c can violate sum-freeness only if

$$c = u_k - u_i - u_j \pmod{K}$$

for some triple of labels. There are at most $b^3 < K$ forbidden shifts. Choose an allowed c, and take T to be the representative of $c - 4p$ modulo K. Translation preserves pair-sum uniqueness and gives (5.1). □

This argument permits all primes p. In particular, no prime-distribution theorem in an arithmetic progression is used in the construction.

6. A Sidon union with arbitrary anchor labels

For an odd prime r, define the increasing ruler

$$a_r(h) = 2rh + \langle h^2 \rangle_r, \qquad h \geq 0.$$

For $0 \leq h < r$ it is Sidon, by the proof of Lemma 3.1 with the high and low components interchanged. More explicitly, the residue sums are less than $2r$, so an equality of ruler pair sums first determines those residue sums modulo $2r$, then the integer sum of the indices, and finally their unordered pair in $\mathbb{F}_r$. Also, for every $h \geq 0$,

$$r < a_r(h+1) - a_r(h) < 3r. \tag{6.1}$$

Choose an alphabet and shift T from Proposition 5.1. For arbitrary labels ℓ_h, put

$$z_h = Ka_r(h) + 4p + T + \delta_{\ell_h} + 2p\varepsilon_{\ell_h},$$
$$A_h = pz_h, \qquad B_h = p(Ka_r(h) + 4p + T). \tag{6.2}$$

Thus $A_h = B_h + p\delta_{\ell_h} + \varepsilon_{\ell_h} Q_p$.

Lemma 6.1. *If p, r are odd primes and $0 \le H < r$, then*

$$S = C_p \cup \{A_h : 0 \le h \le H\}$$

is Sidon for every choice of the labels.

Proof. An equality $z_i + z_j = z_k + z_l$ first determines the unordered label pair modulo K. The sums of the label offsets are consequently equal as integers. Subtract them and divide by K to obtain an equality of ruler pair sums. The ruler is Sidon on $[0, r)$, so $\{i, j\} = \{k, l\}$. Moreover, $z_i + z_j = z_k$ is impossible by sum-freeness modulo K.

It remains to check pair sums involving the seed. Every seed mark is less than $2p^2$, while every anchor is a multiple of p and at least $4p^2$. Hence a sum of two seed marks cannot equal a sum involving an anchor. If

$$c_p(t) + A_i = c_p(u) + A_j,$$

reduction modulo p gives $t = u$ and then $A_i = A_j$. Finally, an equality $A_i + A_j = c_p(t) + A_k$ forces $t = 0$ modulo p, so $c_p(t) = 0$ and $z_i + z_j = z_k$, which has already been excluded. These cases exhaust the possible equalities. □

7. A periodic word for all local windows

Fix a covering alphabet $\mathcal{L}$, its bounds b, M, and an integer $K \ge 1$. The baselines B_h in (6.2) do not depend on the labels. We show that a single periodic word works for all arithmetic data.

Proposition 7.1 (Uniform periodic coverage)**.** *There are $R \ge 2$ and a word $w \in \mathcal{L}^R$ such that the labels $\ell_h = w_{h \bmod R}$ have the following property. For every odd prime p, every $r \ge 1$, every $0 \le h < r$, every $T \ge 0$, and every integer x with $p \nmid x$, if*

$$x - B_{h+j} \in I_p(M) \qquad (0 \le j < R), \tag{7.1}$$

then $x \in A_{h+j} + D_p$ for at least one $j \in \{0, \dots, R-1\}$.

7.1. **The carry vector.** For a window starting at h, write

$$e_j = \left\lfloor \frac{\langle h^2 \rangle_r}{r} + 2\frac{h}{r}j + \frac{j^2}{r} \right\rfloor. \tag{7.2}$$

Direct subtraction gives the exact identity

$$a_r(h + j) - a_r(h) = 2(r + h)j + j^2 - re_j. \tag{7.3}$$

Since $h < r$, the three coefficients in (7.2) belong to $[0, 1] \times [0, 2] \times [0, 1]$. For $j < R$, the expression being rounded is between 0 and $(R + 1)^2$. Its floor is determined by $O(R^3)$ affine integer-level tests in three variables. Lemma 2.1 therefore gives $O(R^{24})$ possible carry vectors. The explicit bound used in Lean is

$$\#\{(e_j)_{j<R}\} \le 25 \cdot 10^8 R^{24}. \tag{7.4}$$

7.2. **The exact good-label criterion.** Set

$$A = 2K(r+h), \qquad B = Kr, \qquad t_j = Aj + Kj^2 - Be_j.$$

Then $B_{h+j} = B_h + pt_j$. Write $x - B_h = d + pm_0$, where $d = \langle x \rangle_p \in \{1, \dots, p-1\}$. For each row and label put

$$\begin{aligned} n_{j\ell} &= m_0 - t_j - \delta_\ell, \\ \sigma_{j\ell} &= \langle n_{j\ell} \rangle_2, \qquad H_{j\ell} = \frac{n_{j\ell} + \sigma_{j\ell}}{2}. \end{aligned} \tag{7.5}$$

These are integer quantities. Under (7.1), $0 < H_{j\ell} < p$: the lower bound on $x - B_{h+j}$ gives $m_0 - t_j \geq 2(M+2)$, while its upper bound gives $m_0 - t_j \leq 2p - 3$; use $0 \leq \delta_\ell \leq M$ and $0 \leq \sigma_{j\ell} \leq 1$.

Let

$$T_{j\ell} = \langle (H_{j\ell} + d^2)(2d)^{-1} \rangle_p.$$

The label ℓ is good in row j, meaning

$$x \in B_{h+j} + p\delta_\ell + \varepsilon_\ell Q_p + D_p,$$

if and only if

$$\begin{aligned} (T_{j\ell} < d) &\iff \sigma_{j\ell} = 1, \\ (\langle T_{j\ell}^2 \rangle_p < H_{j\ell}) &\iff \varepsilon_\ell = 1. \end{aligned} \tag{7.6}$$

For sufficiency, apply Lemma 3.2 and (7.5). For necessity, a representation by two seed marks first fixes their parameter difference modulo p, and then fixes $T_{j\ell}$ by the quadratic equation. Their actual parameter difference is d or $d - p$, and their actual quadratic-residue difference is $H_{j\ell}$ or $H_{j\ell} - p$. The bounds on these differences force exactly the two choices in (7.6).

7.3. **Counting the matrices.** For a fixed carry vector, the pattern $\sigma_{j\ell}$ depends only on the three parity bits of m_0, A, B, because K and the alphabet are fixed. There are at most eight parity patterns. Let $\rho = (4d)^{-1} \in \mathbb{F}_p$ and introduce the five integer features

$$F_{j\ell} = (1, j, j^2, e_j, \delta_\ell - \sigma_{j\ell}). \tag{7.7}$$

In $\mathbb{F}_p$, the parameter $T_{j\ell}$ is the linear combination of these features with coefficients

$$\rho(m_0 + 2d^2), \quad -\rho A, \quad -\rho K, \quad \rho B, \quad -\rho.$$

Normalize their least residues by p. The normalized residue $T_{j\ell}/p$ is the fractional part of the resulting real linear form. Its square residue is obtained from the twenty-five ordered products of the five features, using the normalized residues of the products of the corresponding field coefficients.

Permit these five linear coefficients and twenty-five quadratic coefficients to vary independently in $[0, 1]$. In addition, use five unrestricted real coefficients for $H_{j\ell}/p$, and one real parameter for d/p. Thus all good-label matrices are included in a family of fractional-part tests in

$$5 + 25 + 5 + 1 = 36$$

real variables. Allowing independent coefficients only enlarges the family. Importantly, the five height coefficients need not be bounded: Lemma 2.2 bounds only the forms inside fractional parts.

Each feature has absolute value at most

$$L = (R+1)^2 + M + 2 \leq (M+6)R^2 \qquad (R \geq 1).$$

The two forms inside the fractional parts have absolute values at most $5L$ and $25L^2$. With bR row-label positions, Lemma 2.2 bounds the number of matrices, for fixed carries and parity bits, by

$$\left[38\left(4bR(2(25L^2+5L)+1)+1\right)^{37}\right]^2. \tag{7.8}$$

This is $O_{b,M}(R^{370})$. Multiplying by (7.4) and the eight parity choices gives

$$\#\{\text{good-label matrices on an } R\text{-window}\} \le C_{b,M}R^{394}. \tag{7.9}$$

For example, one may take

$$C_{b,M} = 25\cdot 10^8\cdot 8\cdot 1444\,\left[4b\big(2(25(M+6)^2+5(M+6))+1\big)+1\right]^{74}.$$

These estimates retain every equality case in the comparisons.

Proof of Proposition 7.1. Keep only those matrices in the counted family whose rows are all nonempty. Every actual window satisfying (7.1) gives such a matrix, by the covering property (4.1). Lemma 2.3, with all cyclic rotations included, supplies an $R \ge 2$ and a word meeting every retained matrix in every rotation, since

$$C_{b,M}R^{395}(1-1/b)^R < 1$$

for sufficiently large R. Fix this word. The labels on a window starting at h form the rotation beginning at $h \bmod R$. A good entry at which this rotation meets the window matrix gives the asserted exact integer representation. □

8. Placement and maximal completion

Fix an alphabet and its periodic word, with period R. For the moment suppose that p, r are odd primes, $r \le p$, and $0 \le T < K$, and that

$$\begin{gathered} 4(M+K+4) \le p, \qquad 6KRr \le p, \qquad R \le r, \\ 10p^2 \le N, \qquad N < B_{r-1}, \qquad \lfloor N/p \rfloor \le p^2. \end{gathered} \tag{8.1}$$

By (6.1), the baselines are strictly increasing and

$$0 < B_{h+1} - B_h \le \Delta, \qquad \Delta = 3Kpr, \qquad R\Delta \le p^2/2. \tag{8.2}$$

Write $J = 2p(M+2)$. The first condition in (8.1) gives $J \le p^2/2$, and therefore

$$J + R\Delta \le p^2 \le 2p(p-1). \tag{8.3}$$

Choose H as the largest integer in $[0, r-1]$ for which

$$B_H + Q_p + pM + 1 \le N. \tag{8.4}$$

This choice exists and satisfies $R - 1 \le H < r - 1$. Indeed, $B_0 = 4p^2 + pT$, so (8.2) gives

$$B_{R-1} \le 4p^2 + pT + R\Delta.$$

Together with $T < K \le p$, $M \le p$, and (8.1), this implies (8.4) at $R-1$ (even the weaker bound $R\Delta \le p^2$ suffices). The last index $r-1$ fails because $N < B_{r-1}$.

All elements of

$$S = C_p \cup \{A_h : 0 \le h \le H\}$$

lie in $[0, N)$, by (8.4). Lemma 6.1 shows that S is Sidon, and

$$|S| \le p + H + 1 \le p + r \le 2p. \tag{8.5}$$

Lemma 8.1 (Interior points are blocked)**.** *If $x \notin S$, $p \nmid x$, and*

$$B_{R-1} + J \le x < B_H + J,$$

then $S \cup \{x\}$ is not Sidon.

Proof. Let t be the largest index in $[0, H]$ with $B_t + J \le x$. The bounds on x give $R - 1 \le t < H$ and $x < B_{t+1} + J$. For $h = t - R + 1$ and $0 \le j < R$, monotonicity and (8.2) yield

$$J \le x - B_{h+j} < J + R\Delta \le 2p(p-1).$$

Proposition 7.1 gives $x = A_{h+j} + c - d$ with $c, d \in C_p$. Hence $x + d = A_{h+j} + c$ is an equality of pair sums in $S \cup \{x\}$. It cannot be a trivial equality of unordered pairs because $x \notin S$. □

The unblocked points are thus contained in

$$[0, B_{R-1} + J] \ \cup \ [B_H + J, N] \ \cup \ p\mathbb{Z}. \tag{8.6}$$

The first interval has diameter at most $10p^2$. The second does too. For the latter assertion, maximality of H gives

$$N < B_{H+1} + Q_p + pM + 1 \le B_H + \Delta + Q_p + pM + 1,$$

and therefore $\max(0, N - B_H - J) \le 10p^2$. The constants in these boundary bounds are intentionally loose.

Lemma 8.2 (Cost of completion)**.** *Under* (8.1)*, some maximal Sidon subset of $[0, N)$ containing S has at most $14p$ elements.*

Proof. Extend S to an inclusion-maximal Sidon subset A of the finite host. Any point blocked by S remains blocked by every superset, so all new marks lie in (8.6).

If a Sidon set has m elements in an interval of diameter L, its $m(m-1)$ ordered nonzero differences are distinct. In particular,

$$m(m-1) \le 2L + 1.$$

For $L \le 10p^2$, this implies $m \le 5p$. For marks divisible by p, divide all marks by p. The resulting set remains Sidon and has diameter at most $\lfloor N/p \rfloor \le p^2$, so it has at most $2p$ elements. Restricting A to each exceptional set preserves Sidonicity. Using these three bounds, allowing overlap, and (8.5), gives

$$|A| \le |S| + 5p + 5p + 2p \le 14p.$$

Translation by 1 gives a maximal Sidon subset of $[1, N]$ with the same cardinality. In the certificate this last step is implemented with a corresponding translation of the exceptional residue class. □

9. Prime selection and proof of the main theorem

Fix the two alphabets and modulus K from Proposition 5.1. Apply Proposition 7.1 to each alphabet with this same K, obtaining periods R_1, R_2. Put

$$\begin{aligned} R_* &= R_1 + R_2, \qquad M_* = M_1 + M_2, \\ L_* &= 12KR_* + 16, \\ n_0 &= 40L_*^2 + 4(M_* + K + 4) + R_* + 2. \end{aligned} \tag{9.1}$$

These are absolute finite integers fixed independently of N.

Proof of Theorem 1.1. Let $N \geq n_0^3$ and choose $n \geq n_0$ such that

$$n^3 \leq N < (n+1)^3.$$

Bertrand's postulate supplies primes

$$L_* n < p \leq 2L_* n, \qquad n < r \leq 2n. \tag{9.2}$$

Both primes are odd. Proposition 5.1 selects a compatible alphabet for this p and a shift $T < K$; use its period $R \leq R_*$ and bound $M \leq M_*$.

We verify (8.1). Since $n \geq n_0$,

$$4(M+K+4) \leq n < p, \qquad R \leq n < r \leq p,$$

and

$$6KRr \leq 12KR_* n \leq L_* n < p.$$

Also,

$$10p^2 \leq 40L_*^2 n^2 \leq n^3 \leq N.$$

For $r \geq 2$, $a_r(r-1) \geq r^2$. Therefore

$$B_{r-1} \geq pKr^2 \geq 16n^3 > (n+1)^3 > N,$$

where $n \geq 2$ and $K \geq 1$. Finally, $p \geq n+1$ gives $N < p^3$, hence $\lfloor N/p \rfloor \leq p^2$.

Lemma 8.2 now provides a maximal Sidon set in $[1, N]$ of cardinality at most

$$14p \leq 28L_* n \leq 28L_* N^{1/3}.$$

Thus one may take $C = 28L_*$ and $N_0 = n_0^3$. □

Remark 9.1. The proof is existential. The finite pattern arguments determine absolute alphabets and words, but the resulting constants have not been optimized. No numerical search for these objects is used in establishing the theorem.

Appendix A. The Lean certificate

The companion file `Erdos156.lean` contains all construction-specific auxiliary results and the theorem `Bounty.target`. It has exactly the three imports required by the challenge:

```
import Mathlib
import FormalConjectures.ErdosProblems.«156»
import TaskSupport
```

The final declaration is

```
namespace Bounty

theorem target : fcTypeOfName% "Erdos156.erdos_156" :=
  target_iff_small_maximal_sets.mpr
    Construction.eventual_small_maximal_sets

end Bounty
```

The target type was checked against the explicit proposition

$$\text{True} \iff (N \mapsto (\text{minMaximalSidonSet}(N) : \mathbb{R})) = O_{N\to\infty}(N \mapsto (N : \mathbb{R})^{1/3}).$$

The reduction `target_iff_small_maximal_sets` identifies this statement with the existence of eventually bounded maximal sets. The proof uses the construction above to supply these sets.

Verification record. The delivered source was compiled in Lean `4.35.0-rc2` with mathlib commit

`065356127b1dc0016f66b7283ce0ce2c4055aa55.`

The challenge bundle is pinned to commit

`bf40872fe09f3c9c94e5b1480823a0622cc1698a.`

Compilation of the single-file deliverable finished with zero errors and zero warnings. All 167 public theorems were audited for axiom dependencies. The final theorem depends exactly on `propext`, `Classical.choice`, and `Quot.sound`; the file contains no admitted proofs or additional axiom declarations. The source file has 3,725 lines and SHA-256 digest

`e266058ad3e0e487ea88440470c6d8dbc0`
`f51999b08daa68c0fce5bd3bdc664b.`

The bibliography includes the standard references for Lean and mathlib [5, 6]. The certificate concerns the precise formal target and the construction lemmas below; it is distinct from editorial or peer review of this manuscript.

Correspondence between the paper and the certificate. In the table below, names beginning with `SignPatterns` are in the namespace `Bounty.SignPatterns`; names beginning with `Window` are in `Bounty.Window`. The later construction is in `Bounty.Construction`.

Mathematical component	Principal Lean declarations
Affine and fractional-part pattern bounds	`SignPatterns.affine_sign_determined_card_le`; `SignPatterns.fractional_phase_card_le`
Finite selection, including rotations	`Bounty.exists_hitting_word_permutations`; `Bounty.exists_hitting_word_of_polynomial_bound`
Quadratic seed and exact carries	`Bounty.quadraticSeed_isSidon`; `Bounty.quadratic_difference_canonical`
Separated covering alphabets	`Construction.exists_separatedAlphabet`
Tags for every prime	`Construction.one_of_two_tag_unique`; `Construction.exists_two_tagged_alphabets`
Uniform window counts and periodic word	`Window.floorFamily_card_polynomial`; `Window.allWindowFamily_card_polynomial`; `Window.CoverAlphabet.exists_ruler_word`
Sidon union and interior blocking	`Construction.seed_anchor_union_isSidon`; `Construction.CoverScheme.blocks_interior`
Truncation and maximal completion	`Construction.CoverScheme.exists_truncation`; `Construction.maximal_completion_shifted`
Prime selection and asymptotic bound	`Construction.choose_primes_at_scale`; `Construction.eventual_small_maximal_sets`; `Bounty.target`

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